

Determination of Viscous Forces on Oscillating Measuring Tubes Part I : Verification of Numerical Tools

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1 Introduction

Flow driven by oscillatory walls is a classical problem of fluid mechanics. Characteristics of an oscillatory flow are often required to be evaluated in several industrial applications such as measurement of bulk viscosity of fluids. Presently Coriolis flow rate meters of Endress+Hauser Flowtec AG are used in order to measure viscosities of various fluids on the basis of viscous force exerted by fluids on measuring tubes of the Coriolis meters. In order to measure viscosity of a fluid, oscillatory motion of the tubes is induced and the viscous force on the tube is measured by an electronic sensor. Presently available numerical tools can be quite helpful in order to understand and characterize the flow and its interaction with the tube walls and moreover can contribute to the optimal design of Coriolis devices. Taking this in account, the present report aims at the validation of the numerical tools (CFD) to simulate fluid motion and its corresponding interactions with structure of tube walls during oscillatory motions.

For this purpose, initially a few classical examples of oscillatory motion has been considered and the numerical solution is verified with the help of existing analytical results. The knowledge gained from these numerical simulations will be used to optimize the actual Coriolis devices i.e. the Promass series in a way that the exact forces on the tubes as a function of displacement or vice versa can be computed.

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2 Verification Cases

For the verification of the numerical solvers, a few test cases of oscillatory motion are considered. Firstly, a case of fluid flow over an oscillatory flat plate is considered. Thereafter, the case of oscillatory round tube has also been taken into account and the results obtained from the present numerical simulations have been compared with the respective analytical results.

2.1 Oscillatory Flat Plate

The oscillatory flat plate over a still fluid is known as Stoke's second problem where one generally not interested in the complete solution but only the periodic steady state solution. For this problem, initially a semi-infinite body of fluid and is bounded by

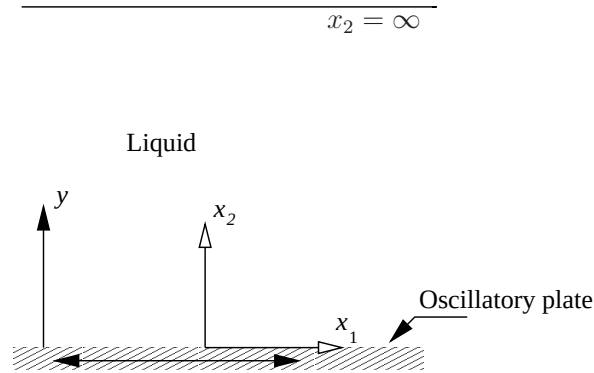


Figure 1: A schematic view of the flow over an oscillatory plate.

a flat plate as shown in Fig. 1 and both fluid and the flat plate are at rest at $t = 0$. Consequently, the oscillatory motion of the flat plate in x direction is initiated such that the plate is displaced by an amplitude A_0 and an angular frequency of $\omega = 2\pi f$. That is the displacement x and velocity U of the plate are given by:

$$\begin{aligned} x(t) &= x_0 \sin(\omega t) \\ U(x_2 = 0, t) &= A_0 \omega \cos(\omega t) = U_0 \cos(\omega t) \end{aligned} \quad (1)$$

where f denotes the frequency of oscillations.

The analytical solution of this is obtained [1, 2] by integrating the following equation for unsteady-state flow:

$$\frac{\partial U}{\partial t} = \nu \frac{\partial^2 U}{\partial x_2^2} \quad (2)$$

On non-dimensionalizing the above equation one can clearly see the influencing parameters of the problem which form the basis to characterize such a problem. Normalizing

U , t , x_2 with U_0 , $1/\omega$ and y one obtains:

$$\frac{\partial U^*}{\partial t^*} = \frac{\nu/\omega}{y^2} \frac{\partial^2 U^*}{\partial x_2^{*2}} \quad (3)$$

where $U = U^*U_0$, $t = t^*/\omega$, $x_2 = x^*y$ and y denotes the dimensional distance from the flat plate. It can be noted that the solution of the above partial differential equation is a function of the dimensionless coefficient of the diffusive term (right hand side term) of eq. (3), i.e. $St = \frac{y^2}{\nu/\omega}$, where St is the Stokes number. From the similarity analysis one can readily learn that for a well spatially resolved CFD simulation, the numerical Stokes number should be less than one, i.e.:

$$\begin{aligned} St_{\text{num}} &= \frac{\Delta x_2^2}{\nu/\omega} < 1 \\ \Rightarrow \Delta x_2 &< \sqrt{\frac{\nu}{\omega}} \end{aligned} \quad (4)$$

which means that the thickness of the cell next to the wall should be lesser than the magic ratio $\sqrt{\nu/\omega}$. Further, the domain length in x_2 direction is given by

$$y \gg \sqrt{\frac{\nu}{\omega}}. \quad (5)$$

It can also be noted from the above analysis that the characteristic distance for an oscillatory problem is in fact scaled with $\sqrt{\nu/\omega}$. That is the propagation distance of the oscillatory motion of the wall increases with increasing viscosity and decreasing frequency and vice-versa.

To integrate eq. (2) for all $t > 0$, the following boundary conditions (B.C.) are employed:

$$\begin{aligned} U &= U_0 \sin(\omega t) \quad \text{at} \quad x_2 = 0 \\ U &= 0 \quad \text{at} \quad x_2 = \infty \gg \sqrt{\nu/\omega} \end{aligned} \quad (6)$$

For the initial condition, the whole system is considered to be stationary, i.e. for $t \leq 0$ and $U = 0$. Finally, on integrating eq. (2) one gets a solution [1]:

$$U = U_0 e^{-\sqrt{\frac{St}{2}}} \cos(\omega t - \sqrt{St/2}) \quad (7)$$

In the argument of the cosine function, the dimensionless quantity “ $-\sqrt{St/2}$ ” is a measure of phase shift between the fluid motion and the oscillatory plate.

Now the same problem is solved numerically for the ratio ν/ω equal to be 10^{-3} . The oscillation frequency was set equal to one, so that the length scale $\sqrt{\nu/\omega}$ can directly be varied via kinematic viscosity only. The numerical grid, as shown in Fig. 2, is constructed such that the for first few cell near the plate

$$\delta y \approx \sqrt{10^{-3}} \approx 0.02 \text{ m}. \quad (8)$$

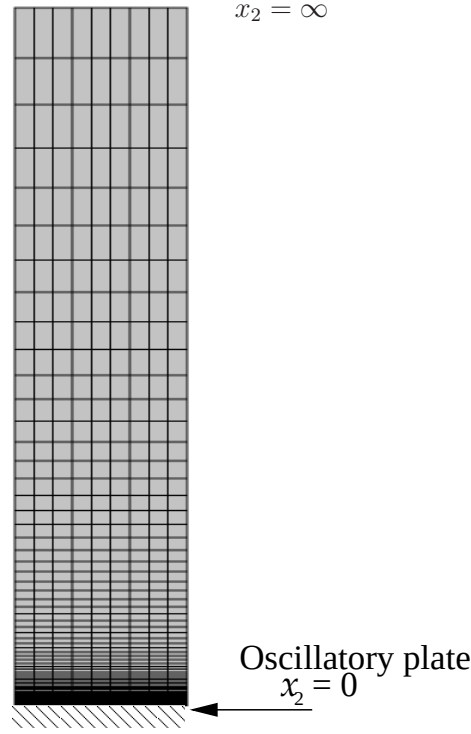


Figure 2: A side view of the numerical grid.

The height of the fluid above the plate was taken to be $4 \text{ m} \gg \sqrt{\nu/\omega}$. The problem is in fact a one-dimensional problem, i.e. the velocity varies only in x_2 direction. Therefore in all other direction either symmetry or periodic boundary conditions are employed.

For the numerical simulations each cycle contains ten steps and the results obtain at the end of the cycle are plotted in Fig. 3. A reasonably good agreement can be found between the analytical and numerical results.

2.2 Oscillatory Tube

The present test case deals with the circular tube which is oscillating around its axis in the azimuthal direction in order to measure the viscosity of the fluid by the effect of shear force exerted by the fluid on its walls. A schematic view of the problem is shown in Fig. 4 where the tube with radius $R = 8.8 \text{ mm}$ oscillates with a frequency $f = 725 \text{ Hz}$. The pipe oscillates in the tangential (θ) direction with an amplitude angle of $\Theta_0 = 1^\circ$. For such a configuration, the velocity of the oscillating tube is prescribed as a sine function, i.e.:

$$U_t(r = R, t) = U_0 \sin(\omega t) = R\Theta_0\omega \sin(\omega t) \quad (9)$$

where ω is the angular frequency of the oscillations, r the radial coordinate direction and U_t the tangential velocity in the azimuthal direction θ .

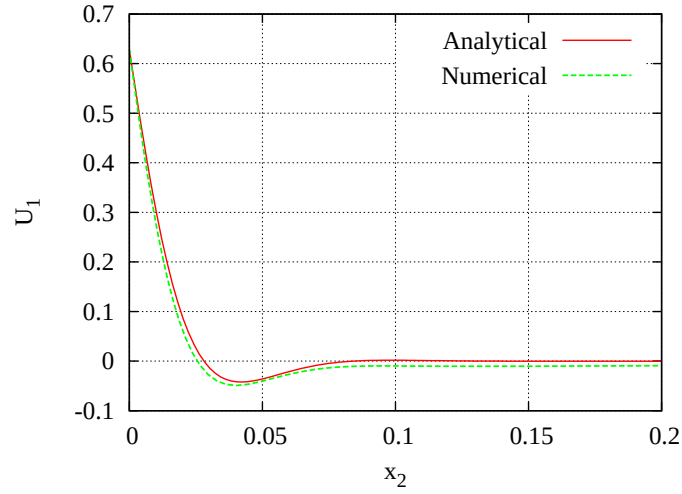


Figure 3: Comparison of periodic numerical and analytical (eq. (7)) solutions for the case of flow induced by oscillatory flat plate underneath a semi-infinite fluid at the end of a cycle.

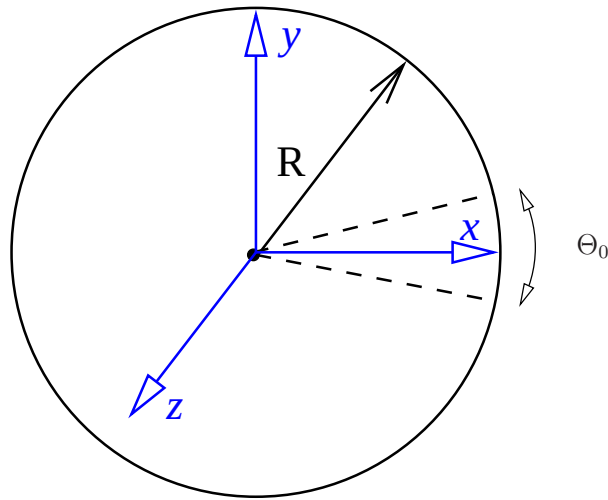


Figure 4: A schematic cross-sectional view of the oscillating around z axis.

An analytical periodic solution of the problem was derived in an earlier report [3] which has the form:

$$U_t(r, t) = \Re\{AJ_1(ar)\} \quad (10)$$

$$\text{with } A = \frac{-i\omega r\Theta_0}{J_1(ar)} \text{ and } a = \sqrt{\frac{i\omega}{\nu}} \quad (11)$$

where $\Re\{g\}$ represents the real part of function g and $J_1(x)$ is a Bessel function of type 1 of first order.

For the numerical solution of the problem only a 2D slice (in x - y plane) of the pipe is considered to be the computational domain. This is due to the fact that the problem is physically two-dimensional (however conceptually it will be still a one-dimensional problem since $\partial/\partial\theta = 0$). Moreover this simplification significantly reduces the computational time. As discussed earlier, the spatial resolution requirement in a numerical computation depends on term $\sqrt{\nu/\omega}$, however, for the temporal resolution there is no clear indication. In fact, it is absolutely a numerical issue than a physical one. It is obvious that the numerical time step should be much smaller than physical time step (which here corresponds to frequency of oscillations, i.e. $1/f$). Therefore, just to start with computations, a time step of one eighth of a cycle ($2\pi/8$) was chosen.

A comparison of the computed and analytically obtained velocities for $\nu = 1000$ cSt is shown in Fig. 5. It can be observed from the figure that agreement between the numerical computations and the analytical solution is fairly good. Since in the problem the velocities of the fluid undergo maxima (at multiple of $\pi/2$) and minima (at multiple of π), we must analyze and evaluate results at these two periodic locations. These two periodic locations are also referred to as imaginary and real part of the solution, respectively.

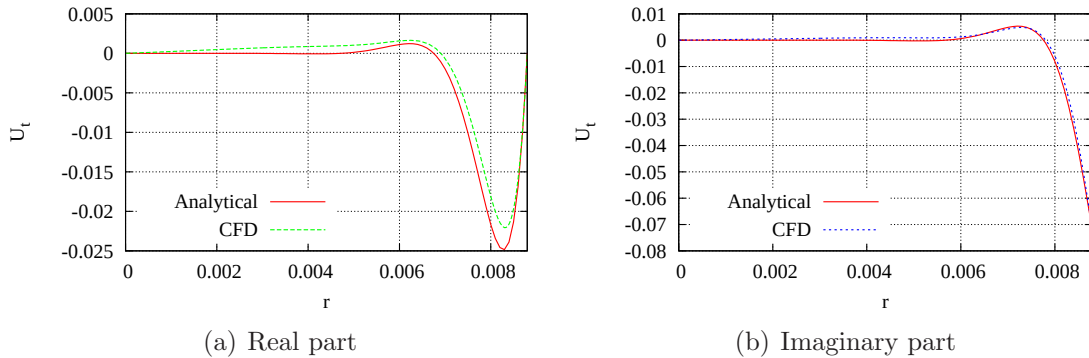


Figure 5: Periodic solution (tangential velocity profile) of the oscillating (in x - y plane) circular tube with 8 time steps per cycle for $\nu = 1000$ cSt.

For the measurement of the viscosity of a fluid, it is important to accurately estimate the forces acting on the oscillating tube. The applied average force by a fluid is directly related to the shear stress exerted on the wall. Therefore, it is important to compare the

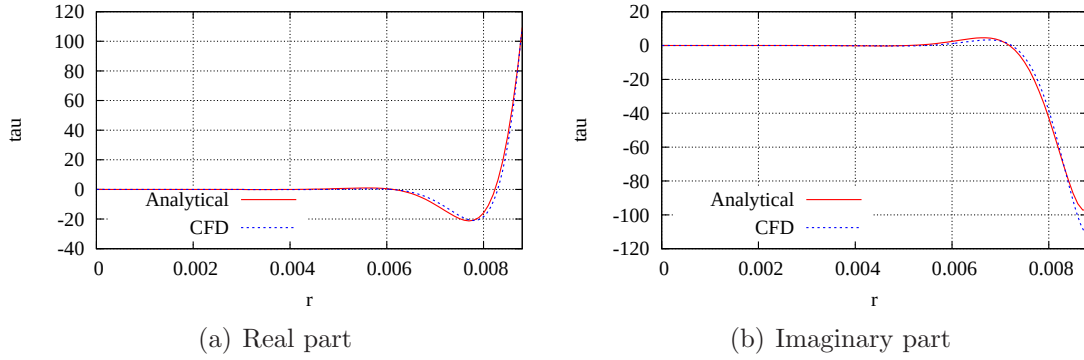


Figure 6: Periodic solution (tangential velocity profile) of the oscillating (in x - y plane) circular tube with 8 time steps per cycle for $\nu = 1000$ cSt.

numerically and analytically obtained values of the shear stress in the domain. As it can be observed from Fig. 6 that though the computed tangential velocity profiles for both real and imaginary part seem to agree well with the analytical results, the radial component of the gradient i.e. $\partial U_t / \partial r$, especially for the imaginary part deviate from the analytical solution in the region close to the wall. These inconsistent numerical results lead to large discrepancy in the computed *wall shear stress* for the imaginary part or at the maxima of the cycle. This is clearly indicated in Fig. 7 which shows a change in slopes of the shear rate curves for both the real and imaginary parts. The imaginary part of the shear stress (Fig. 7) shows an inversion when the viscosity is increased over a certain value. This inversion point as well as the slope of the shear stress curve is not accurately predicted with the present temporal resolution.

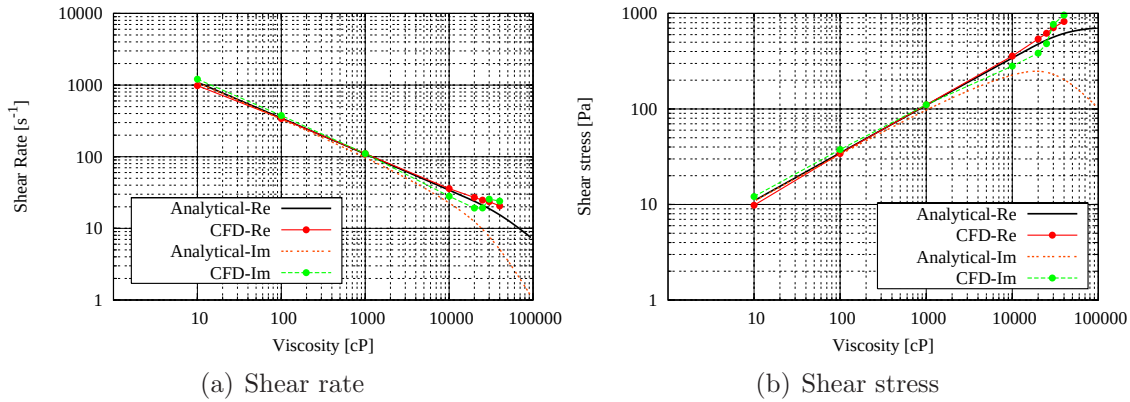


Figure 7: Numerically predicted shear rates and stresses as a function of dynamic viscosity with 8 time steps per cycle.

The reason for this disagreement was that the phase lag between the oscillating wall and fluid adjacent to it could not be well captured due to accuracy of the time marching scheme. Though the second order Euler scheme was adopted for the time integration, *eight time steps seem to be inadequate* for accurately resolving the time response of the

fluid. Taking this consideration in account, the number of time step within a cycle were increased to 2^n where $n = 4, 5, \dots, 9$ for the case with $\nu = 20000$ cSt. The computed values of the radial shear stress, with increased temporal resolution is plotted in Fig. 8. It can be observed from the figure that the real part of the stress is more or less same

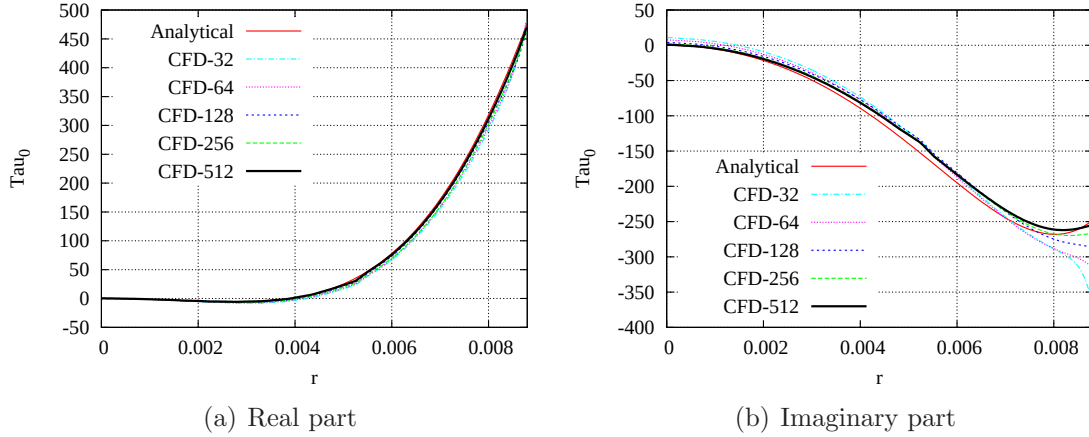


Figure 8: A comparison of the computed radial stresses with the analytical results with increasing temporal resolution as 2^n where $n = 5, 6, \dots, 9$ for $\nu = 20000$ cSt.

for all the cases. However, the computed imaginary part indicate much improvement with increased temporal resolution. It is interesting to note that the change in sign of the second derivative of the velocity close the wall is not at all captured with temporal resolution lower than 256. This seems to be a serious limitation of a commercial CFD program like CFX where the maximum accuracy of the time integration is usually of second order. Therefore, from the present analysis it has been realized that for a reasonably accurate analysis one requires at least 256 time steps in a period especially near the point where the slope of the shear stress versus viscosity curve (Fig. 9) changes. This point is known can be termed as inversion or in-flex point. The position of the inversion in shear stress depends on the oscillation frequency and the length scale i.e. the diameter of the pipe. It has already been shown in [3] that

$$\nu_{\text{inv}} \sim R^2 f \quad (12)$$

where the subscript “inv” indicate the *inversion point* and R is the radius of the pipe and f the oscillating frequency. From the above relation one can readily conclude that the for really small frequencies and pipe diameters, the inversion point will be at relatively lower viscosity and consequently the numerical accuracy may need to be taken care even at much lower viscosities unlike in the present exercise where the inversion takes place at around 20000 cSt.

As shown earlier that with 8 numerical steps per cycle do not predict the inversion point correctly, simulations for different viscosity fluids are repeated by increasing the temporal resolution to 512. Figure 9 shows the comparison between the numerical and analytical results. The high temporal resolution computations indicate a much better agreement with the analytical results.

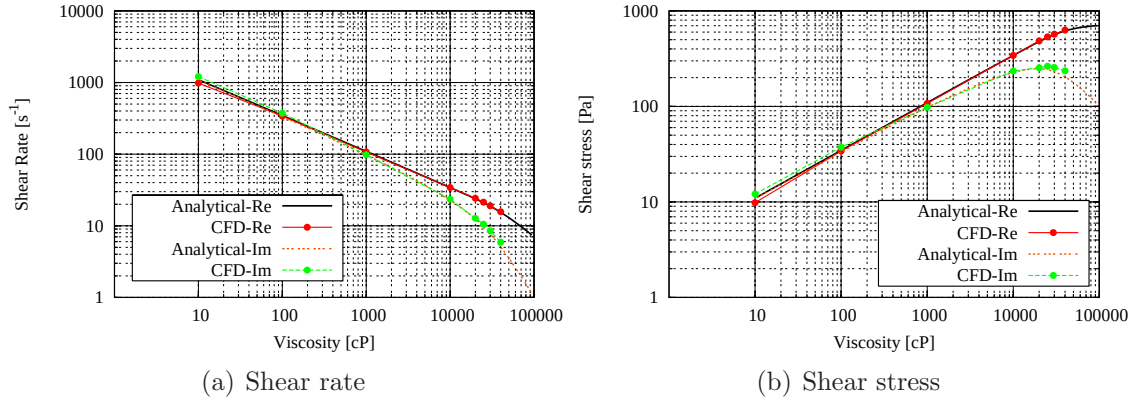


Figure 9: Numerically predicted shear rates and stresses as a function of dynamic viscosity with 512 time steps per cycle.

3 Conclusions

The present numerical work was carried out in order to verify the numerical simulations with the existing analytical solutions in the presence of oscillatory walls. It has been realized that for accurate numerical predictions, one requires to resolve the velocity gradients close to the wall on the basis of ratio $\sqrt{\nu/\omega}$. Furthermore, numerical simulations require fairly high temporal resolution in order to accurately resolve the imaginary part of the shear stress or to well define complete periodic behavior.

References

- [1] R.B. Bird, W.E. Stewart, and E.N. Lightfoot. *Transport Phenomena*. John Wiley & Sons, 1960.
- [2] F.M. White. *Viscous Fluid Flow*. McGraw Hill Inc., 2 edition, 1991.
- [3] Christof Huber. Theorie der Viscositätsmessung mit einem Coriolis-Masse-Durchfluss-Sensor. Internal report, Endress+Hauser Flowtec AG, 2006.